WORKED SOLUTION

STEP III 2002, Question 10

Mechanics • Friction

STEP III 2002, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q10

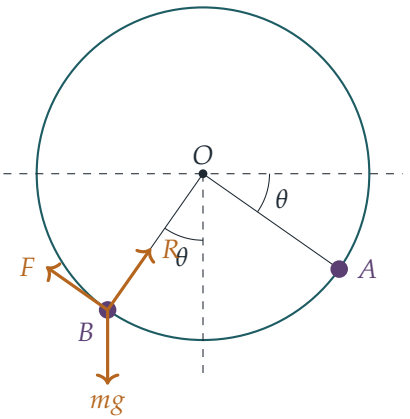
The key idea

As long as the second particle does not slip, it moves with the cylinder, so the whole system is a rigid body turning about a fixed axis: two equal masses at radius a, a quarter-turn apart. Energy (or moments about the axis) then gives the motion in one line. To find when slipping starts, we look at the loose particle on its own and use Newton's second law along and across the surface to find the friction F and the normal reaction R it needs. It slips when F first reaches μR.

Solution

Setting up

Model both particles as point masses on a light cylinder, with a smooth axle. Let O be the axis, A the fixed particle and B the loose one. Released from rest, A falls, so the cylinder turns (clockwise in the figure) and carries B up the side away from A. After a turn through θ, OA is at angle θ below the horizontal and OB is at angle θ to the downward vertical, on the other side.



The equation of motion

While B does not slip, both particles move on circles of radius a with angular speed $\dot\theta$, and the cylinder itself has no mass. Taking the level of O as zero potential energy, A is at height $-a\sin\theta$ and B at height $-a\cos\theta$. The system starts from rest at $\theta = 0$, and no work is done by the smooth axle, while friction between B and the cylinder does no net work because there is no relative motion. So energy is conserved:

$$2\cdot\tfrac{1}{2}ma^2\dot\theta^2 - mga\sin\theta - mga\cos\theta = -mga,$$

that is,

$$\dot\theta^2 = \frac{g}{a}\,(\sin\theta + \cos\theta - 1). \qquad (*)$$

Differentiating with respect to time and dividing by $2\dot\theta$ (which is non-zero once motion has started) gives

$$\ddot\theta = \frac{g}{2a}\,(\cos\theta - \sin\theta),$$

as required. Equivalently, taking moments about O: the weight of A has moment $mga\cos\theta$ in the direction of rotation, that of B has moment $mga\sin\theta$ against it, and the moment of inertia is $2ma^2$.

When does B slip?

Now look at B alone. The forces on it are its weight mg, the normal reaction R towards O, and friction F along the surface, which we take as positive in B's direction of motion (up the side). B moves on a circle of radius a, so it has acceleration $a\dot\theta^2$ towards O and $a\ddot\theta$ along its direction of motion. The weight has component $mg\cos\theta$ directly away from O and $mg\sin\theta$ against the motion. Hence

$$\begin{aligned}\text{towards } O: &\quad R - mg\cos\theta = ma\dot\theta^2,\\ \text{along the motion:} &\quad F - mg\sin\theta = ma\ddot\theta.\end{aligned}$$

Using $(*)$ and the equation of motion,

$$\begin{aligned}R &= mg\cos\theta + mg(\sin\theta + \cos\theta - 1) = mg\,(2\cos\theta + \sin\theta - 1),\\ F &= mg\sin\theta + \tfrac{1}{2}mg(\cos\theta - \sin\theta) = \tfrac{1}{2}mg\,(\sin\theta + \cos\theta).\end{aligned}$$

For $0 \le \theta \le \frac{\pi}{3}$ both are positive. ($R > 0$ because $2\cos\theta + \sin\theta - 1 \ge \sin\theta > 0$ for $0 < \theta \le \frac{\pi}{3}$, using $2\cos\theta \ge 1$, and $R = mg$ at $\theta = 0$.) So B stays in contact, and friction pushes it up the side. The particle does not slip as long as $F \le \mu R$, that is, as long as

$$\rho(\theta) = \frac{F}{R} = \frac{\sin\theta + \cos\theta}{2(2\cos\theta + \sin\theta - 1)} \le \mu.$$

At $\theta = \frac{\pi}{3}$, where $\sin\theta = \frac{\sqrt3}{2}$ and $\cos\theta = \frac{1}{2}$,

$$\rho\left(\tfrac{\pi}{3}\right) = \frac{\frac{1}{2}(\sqrt3+1)}{2\cdot\frac{\sqrt3}{2}} = \frac{\sqrt3+1}{2\sqrt3} = \frac{3+\sqrt3}{6} = \mu.$$

Ask yourself Why can't the particle have started to slip earlier than this?

We need $\rho(\theta) < \mu$ for all $0 \le \theta < \frac{\pi}{3}$. Write $N = \sin\theta + \cos\theta$ and $D = 2\cos\theta + \sin\theta - 1$. Then

$$\begin{aligned} N'D - ND' &= (\cos\theta - \sin\theta)(2\cos\theta + \sin\theta - 1) - (\sin\theta + \cos\theta)(\cos\theta - 2\sin\theta) \\ &= \cos^2\theta + \sin^2\theta + \sin\theta - \cos\theta = 1 + \sin\theta - \cos\theta, \end{aligned}$$

which is positive for $0 < \theta < \pi$. Since $D > 0$ on $[0, \frac{\pi}{3}]$, $\rho' = (N'D - ND')/(2D^2) > 0$ there, so ρ is strictly increasing from $\rho(0) = \frac{1}{2}$. It therefore stays below μ until it reaches μ exactly at $\theta = \frac{\pi}{3}$. Also, by $(*)$, $\dot\theta^2 > 0$ for $0 < \theta < \frac{\pi}{2}$, so the cylinder really does keep turning and reaches this angle.

So the second particle starts to slip when the cylinder has turned through $60°$, as required.

Checks

- At $\theta = 0$: B is at rest at the bottom, yet $F = \frac{1}{2}mg \neq 0$. That is right: A's weight is trying to turn the cylinder, and friction must accelerate B along with it. Indeed $F = ma\ddot\theta = \frac{1}{2}mg$ there, and $R = mg$ because B has no speed yet.
- At $\theta = \frac{\pi}{4}$, $\ddot\theta = 0$ and the angular speed is a maximum, as expected when the two weights balance about O.
- Numerically, integrating the equation of motion and computing the contact force on B directly from its acceleration reproduces the formulas for R and F, and F/R first reaches $(3 + \sqrt{3})/6 \approx 0.7887$ at $60.00°$.

Common slips

- Including a moment of inertia for the cylinder: it is light, so only the two particles contribute $2ma^2$.
- Forgetting the tangential acceleration $a\ddot\theta$ of B, which would give $F = mg\sin\theta$ and the wrong angle.
- Checking only that $F = \mu R$ at $60°$ without showing that F/R was smaller before then.

Beyond STEP

This is a one-degree-of-freedom system: once the constraint (no slipping) is imposed, a single coordinate θ describes everything, and energy gives the motion directly. The constraint force is found afterwards, and the constraint fails when the force it needs exceeds what the surface can supply. This is the same logic as a particle leaving a

sphere, and in Lagrangian mechanics it is made systematic with Lagrange multipliers. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops rigid-body rotation and constraint forces in this spirit.

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